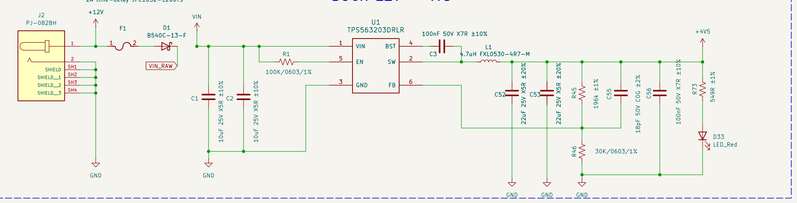
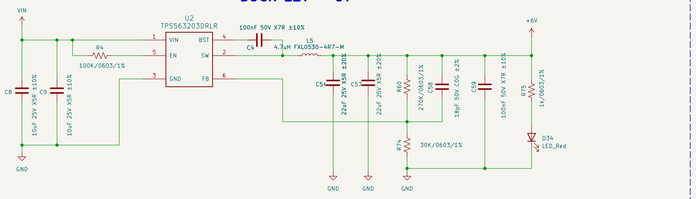
1_Power

2A time-delay JFC1032-1200TS

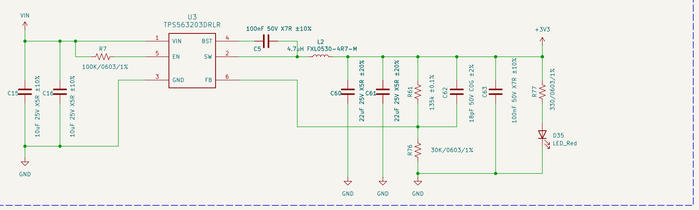
BUCK 12V – 4V5



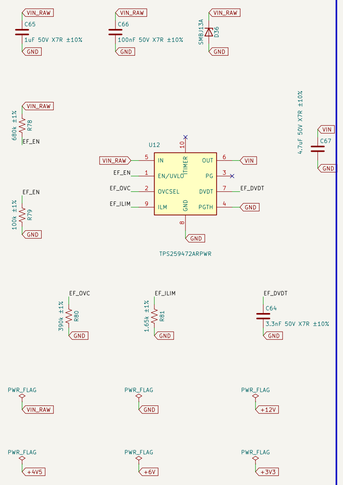
BUCK 12V – 6V



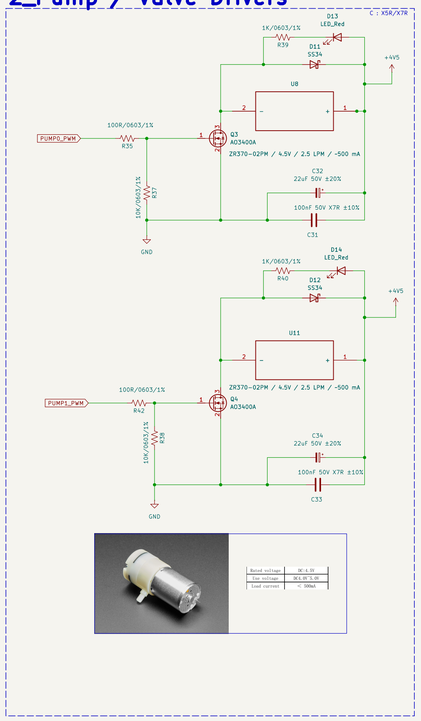
BUCK 12V – 3V3



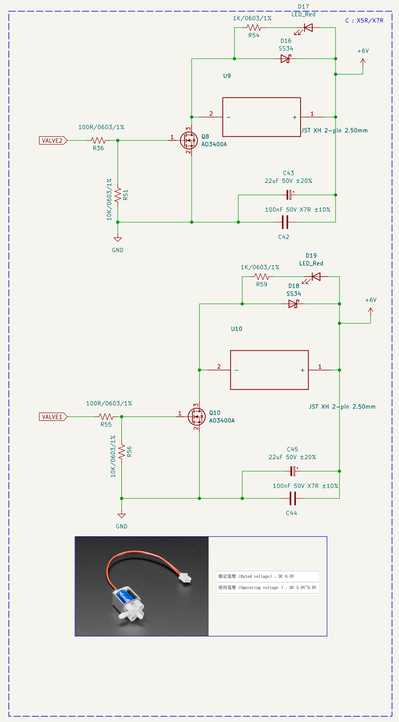
INPUT PROTECTION — REGULATED 12 V ONLY
13.8 V clamp; ~2 A current limit; ~10 ms soft start
Validate startup, hot plug and simultaneous loads.



2_Pump / Valve Drivers

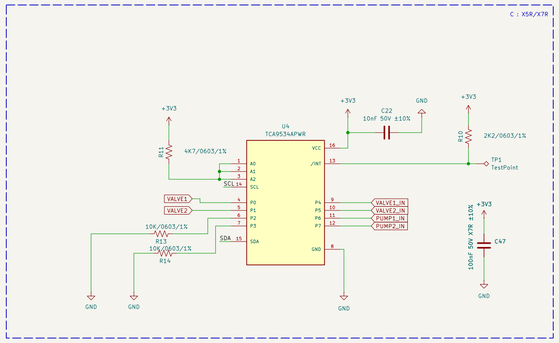


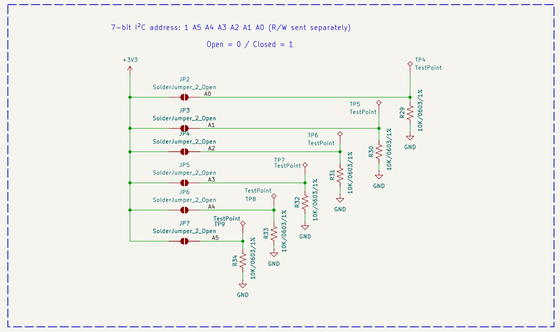
Rated voltage	DC 6.0V
Use voltage	DC 5.0V~7.0V
Load current	< 500mA

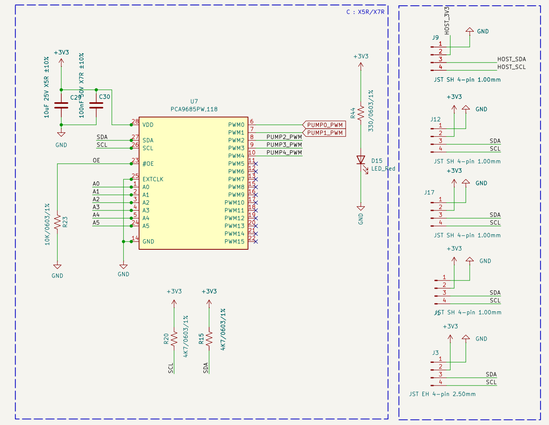


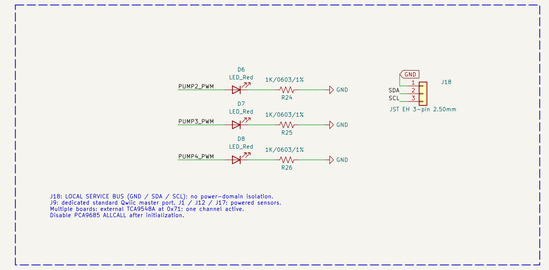
额定电压 (Rated voltage)	DC 6.0V
使用电压 (Operating voltage)	DC 5.0V~6.0V

3_PERIPHERAL



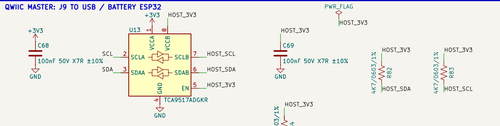






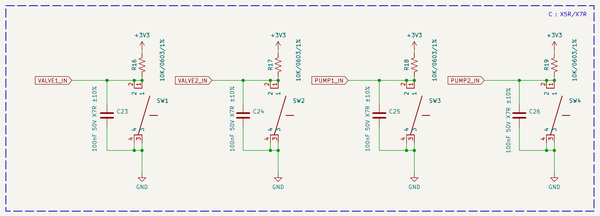
J18: LOCAL SERVICE BUS (GND / SDA / SCL): no power-domain isolation.
J9: dedicated standard Qwiic master port, J1 / J12 / J17: powered sensors.
Multiple boards: external TCA9548A at 0x71; one channel active.
Disable PCA9685 ALLCALL after initialization.

QWIIC MASTER: J9 TO USB / BATTERY ESP32



LOCAL A: RheoBoard +3V3 and sensor bus
HOST B: ESP32-supplied +3V3; regulators never paralleled
100 kHz initial bring-up; cable/module pull-ups share a current budget
J1 / J12 / J17 remain powered Qwiic sensor ports
J18 = LOCAL BUS SERVICE; no master-power isolation at that header

4_Switch Inputs



5_Pressure Sensor

